

ELECTRONICS IN CARS

Besides music systems and LCD screens, is there any significant role that electronics plays when it comes to cars? That was a question asked by a reader—here we attempt to answer it



Bosch technology for the future of mobility. Protecting the environment and preventing accidents are the main issues

■ ABHIJIT PAUL CHOUDHURY

Necessity is the mother of invention. If you peep into the history, you would realise that cars were originally made

of only mechanical parts, but today electronics constitute a major part. Be it the engine, the dashboard or any other part, the use of electronics can be found. But, why was electronics used in cars? Definitely, to make them more efficient, simple and perform better.

The overall trend

Automotive is one of the fastest growing sectors in India. There is a constantly growing requirement for environment protection. Also, the end users are demanding greater fuel efficiency, security and safety. This has greatly contributed to the rapid development of the technology that can be employed in the automobiles. Cars, which used to be only mechanical, now contain more than 1000 electronic components—most of which happen to be sensors and microcontrollers.

What all is going digital?

That's the question we started off with. Of course, we began with the basic stuff like the music system and the digital clock before we started working in-depth on this story. But, then as the information started pouring in, we were left wondering whether there was anything left (in a car) where electronics is not playing a significant role.

The engine. The engine is the heart of a car. The circuit that automates how much fuel should enter the engine is governed by the electronic control unit (ECU). The ECU consists of sensors and microcontrollers, which govern the flow of fuel into the engine. It decides the amount of fuel to be injected inside the engine with the help of the pressure sensor, throttle position sensor, oxygen sensor, fuel injector and many other components. The main aim of using the ECU is to increase the fuel efficiency of a car.

The transmission. Typically, there are two types of transmission used in cars—manual and automatic, also referred to as manual gearing or auto-